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Tribological Performance of Manganese Tetraoxide-Weighted Potassium Formate Brine as Reservoir Drilling Fluid

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Abstract

The depletion of conventional fossil fuel reserves has compelled the oil and gas industry to adopt unconventional drilling techniques, such as deepwater, multilateral, and horizontal drilling. To access previously unreachable reserves, advanced technologies like extended-reach drilling (ERD) are employed. However, these methods often present operational challenges, particularly under high-pressure, high-temperature (HPHT) conditions. A critical factor in addressing these challenges is the selection of appropriate drilling fluids that ensure environmental safety while maintaining operational efficiency. Non-aqueous drilling fluids (NADFs) are often preferred for their superior thermal stability and high rate of penetration; however, they present significant environmental concerns and incur high disposal costs. As a result, there is an interest in high-performance, environmentally friendly alternatives such as potassium formate (KFo) brine-based aqueous fluids, which offer a balance between operational efficiency and environmental compliance. However, its limited maximum saturation density (1.57 SG) restricts its application in deep wells. This limitation can be addressed by adding solid weighting agents. Manganese tetraoxide (Mn_3O_4) offers high density (4.8 SG), and ultra-fine particle size, making it compatible with KFo brines. A challenge with aqueous fluids, especially high-density brines, is their higher friction factor during drilling compared to NADFs, which can hinder operations in deviated wells. Existing lubricants often underperform in saturated brines, prompting the need for tailored solutions. This study evaluates the frictional and wear behavior of cesium and potassium formate brines, as well as Mn_3O_4 /KFo system, in the presence of three lubricant additives aimed at reducing friction. The best-performing additive was further analyzed using Quartz Crystal Microbalance (QCM) techniques to simulate downhole conditions. Final tests with high-density fluids confirmed improved frictional performance, supporting the viability of KFo-based systems as an NADF alternatives.

Introduction

The ongoing exploration of oil and gas resources has led to a decline in the availability of conventional onshore fossil fuels, prompting the industry to expand operations offshore and adopt unconventional extraction methods such as shale oil and gas. Consequently, drilling techniques have evolved from traditional vertical drilling to more complex approaches like extended reach drilling (ERD) [1]. These operational advancements introduce new challenges, and underscore the critical importance of selecting drilling fluids that balance environmental safety with operational performance. The appropriate choice of drilling fluids is essential not only for enhancing drilling efficiency but also for protecting drill bits and preventing drill pipe failures caused by excessive heat and material degradation during the drilling process [2]. For decades, non-aqueous drilling fluids (NADFs) have been the preferred choice for drilling challenging wells due to their robustness, rate of penetration (ROP), and excellent thermal stability. However, there is an interest in developing alternative drilling fluids to replace NADFs, which are both costly and environmentally unfriendly. These alternative drilling fluids must offer an improved health, safety, and environmental (HSE) profile while matching the performance of non-aqueous drilling fluids.

Potassium formate (KFo) brine is used as an aqueous reservoir drilling fluid, particularly in sensitive formations such as shales and hydratable clays. However, its maximum saturation density of 1.57 SG is a limitation. To extend its use beyond this density, solid weighting agents are added to increase the brine's density. Barite, on the other hand, is unsuitable for weighting KFo brine because it might dissolve and forms barium formate, a substance that is toxic and environmentally hazardous.

Manganese tetraoxide (Mn_3O_4) is a versatile weighting material for drilling and completion fluids. It has a high density (4.8 SG), is insoluble in KFo brine but acid-soluble, and consists of micro-spherical particles ($d_{50} \approx 1 \mu\text{m}$), making it suitable for use with KFo brines. Mn_3O_4 /KFo fluids with densities ranging from 1.60 to 2.35 SG have been successfully used in drilling operations worldwide. These fluids are environmentally friendly, provide excellent shale stability, and demonstrate high thermal resilience. Additionally, they exhibit exceptional stability and recyclability, contributing to enhanced drilling performance.

Aqueous drilling fluids, especially saturated brines, are known to have higher friction factors during drilling compared to NADFs. This poses challenges when drilling long or highly deviated wells unless an effective lubricant is added to reduce friction. While many commercially available lubricants perform well in freshwater-based fluids, their performance tends to decline in highly saturated brines. As a result, there is a clear need to develop lubricants specifically tailored for formate brines.

In this study, two different solid-free formate brines, cesium formate (CsF) and potassium formate (KFo), were evaluated for their frictional and wear performance, with KFo demonstrating improved frictional behavior. Following this, an Mn_3O_4 /KFo mixture was formulated, and the impact of three different additives on the drilling fluid's frictional performance was investigated in a steel-steel contact configuration. After identifying the most effective additive, additional tests were conducted using a new configuration (steel-stone), and the results were compared. To gain deeper insights into the mechanisms influencing friction, Quartz Crystal Microbalance (QCM) studies were performed using Fe-coated and Kaolinite-coated sensors, designed to simulate steel and stone surfaces, respectively. Finally, higher-density water-based drilling fluids, which better represent actual drilling conditions, were formulated using the best-performing lubricant, and their frictional performance in steel-stone interactions was assessed.

Experimental setup

Materials and preparation. As mentioned in the previous section, tests have been performed using steel/steel and steel/stone configurations. In the case of steel/steel configuration, both disc and ball were made from SAE52100 bearing steel supplied by Basedo Steel company with a chemical composition mentioned in Table 1. In steel/stone configuration, the same SAE52100 balls were used in contact with stone discs

prepared from shale. The X-ray diffraction pattern (XRD) of the shale sample revealed Kaolinite as the main phase.

Table 1—Chemical composition of SAE52100 bearing steel

	Fe	Cr	Mn	Si	C	P	S
SAE52100	Balance	1.3-1.6	0.25-0.45	0.15-0.3	0.98-1.1	≤0.025	≤0.025

Both steel and stone discs were ground and polished to reach a surface roughness (Ra) of 0.35 μm while the roughness of the balls was 0.05 μm . Both counter surfaces (disc and ball) were ultrasonically cleaned in a 50-50 distilled water-ethanol mixture for 5 minutes.

The QCM measurements were performed using Fe-coated and Kaolinite-coated Quartz crystal sensors to simulate surface reactions on steel and stone, respectively. The coated sensors were supplied by Biolin Scientific and the chemistries were decided based on the chemical composition of the steel and the stone.

Three different water-based fluids were formulated in this study based on Mn_3O_4 /Kfo mixture as presented in Table 2.

Table 2—Formulation of Mn_3O_4 /Kfo fluids

	Fluid 1, (1.85 SG)		Fluid 2, (2.00 SG)		Fluid 3, (2.3 SG)	
Additives in order of addition	Mass [g]	Vol. [mL]	Mass [g]	Vol. [mL]	Mass [g]	Vol. [mL]
Potassium Formate brine	491.0	316.5	452.2	291.5	407.3	262.8
Xanthan gum polymer	-	-	0.3	0.2	-	-
Synthetic polymer	-	-	-	-	2.4	1.6
Cellulose polymer	-	-	-	-	2.0	1.3
HT Starch	-	-	5.0	3.3	-	-
Milled CaCO_3	-	-	10.1	3.7	5.1	1.9
ESMD2 dispersant	-	-	3.5	2.9	1.5	1.3
Defoamer	1.1	1.1	1.0	1.1	0.9	1.0
Sodium bicarbonate	0.5	0.2	0.5	0.2	0.5	0.2
Sodium carbonate	1.1	0.4	1.0	0.4	1.0	0.4
Mn_3O_4	154.2	31.8	226.7	46.7	385.6	79.5
Total	647.9	350.0	700.3	350.0	806.3	350.0

As shown in Table 2, Fluid 1 was formulated with minimal additives to isolate and evaluate the effect of Mn_3O_4 on friction, using pure Kfo brine as a baseline. In contrast, Fluids 2 and 3 incorporated typical drilling fluid additives, including a viscosifier, fluid loss control agents, a dispersant, a silicone-based defoamer, and a carbonate buffer.

Filtration properties are critical parameters in drilling fluid performance. To enhance these properties and minimize formation damage, filtration control additives are commonly employed in water-based drilling fluids. These typically include polymers such as synthetic AMPS-based polymers, cellulose derivatives, and high-temperature (HT) starch [3, 4].

All fluid samples were subjected to heat aging at 135 $^{\circ}\text{C}$ for 16 hours prior to measuring the coefficient of friction (CoF).

Tests and measurements. Three different additives (F, M and S) were selected to study their effect on friction of Mn_3O_4 / KFo fluids. Additive F is anionic polymer, additive M is modified coconut oil laurate and additive S is a glucose based surfactant.

Figure 1 demonstrates the pin-on-disc test principle utilized in this work. In this configuration, a fixed ball meets a rotating disc in a sliding motion, and the upper and lower sample holders are used to mount the samples. The friction and wear tests were conducted on an MFT-5000 multi-function tribometer (Rtec Instruments, USA) using unidirectional sliding motion.

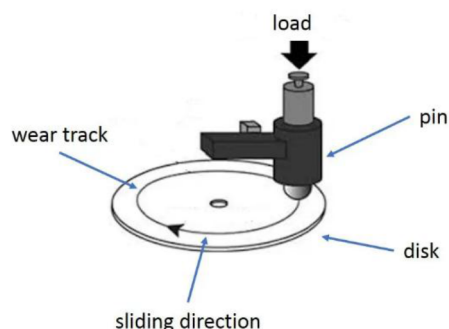


Figure 1—Illustration of the pin-on-disc test configuration

Figure 2 shows the MTF-5000 tribometer setup. The lower stage and upper stage are driven by separate electric motors. Applying a normal load by pressing the ball against the disc, will reveal a tangential force (in X-direction or Y-direction) on the ball while sliding. That force can be measured by the load cells. The coefficient of friction (CoF) is obtained by dividing the tangential measured force by the applied normal load.

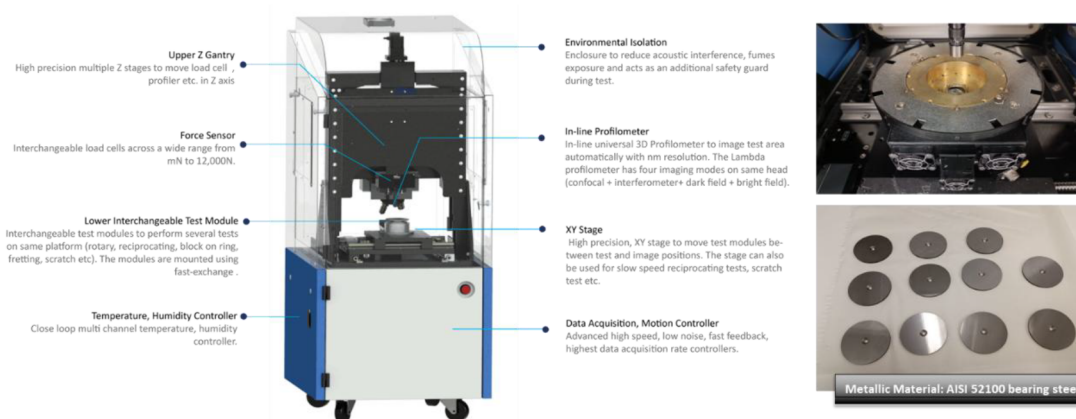


Figure 2—The MFT5000 tribometer (Rtec Instruments, USA)

The testing procedure involved applying a normal load of 10 N, which resulted in an initial maximum Hertzian contact pressure of 1.07 GPa. The tests were conducted at a temperature of 50°C. The rotation speed of the disc was 150 rpm and the track diameter was 50 mm, which resulted in a sliding speed of 0.39 m/s. Tests were repeated at least twice to account for repeatability, in many instances more than 2 tests were necessary however only the results of two tests will be shown in this work. The parameters were chosen to be close to the common set up for oilfield lubrication testing. The worn surfaces (on the disc and the ball) were studied using a 3D microscope located on the back side of the MFT5000 tribometer.

The surface adsorption studies were performed using a Quartz crystal microbalance with dissipation mode (QCM-D) supplied by Biolin Scientific (Sweden). Fe coated and Kaolinite-coated AT-cut 5 MHz

sensors from QSense (Biolin Scientific, Sweden) with a surface roughness less than 1 nm were used. The flow rate was maintained at 0.05 mL/min during the experiments using a peristaltic pump. The resonance frequency and dissipation shift for the fundamental frequency and third, fifth, and seventh overtones were recorded. More details about the QCM-D tests and viscoelastic modeling of the adsorbed layers have been discussed elsewhere [5].

Results and Discussion

Potassium and Cesium formate brines. Wellbore instability in shales is one of the primary causes of non-productive time (NPT) resulting in increased cost during drilling operations. Issues arising from instability pose numerous challenges such as reduced hole cleaning due to reduced annular velocities in enlarged sections of the wellbore, and severe cases such as stuck pipe and pack offs caused by wellbore caving and collapse [6]. While wellbore stability issues can occur with oil-based and synthetic-based muds, they are particularly problematic with water-based muds. The interaction between water-based muds and problematic shales has prompted the development of a wide range of shale inhibitors designed for water-based systems. Some of these additives are tailored to specific shale types, while others are restricted to certain operating conditions. However, while some inhibitors stabilize cuttings effectively, they may have little to no impact—or even a negative effect—on the overall stability of the wellbore.

Among water-based drilling and completion fluids, potassium chloride, potassium and cesium formate brines are considered among the most effective in stabilizing shales. Unlike conventional water-based fluids, the stabilizing properties of formate fluids come from the intrinsic characteristics of the formate brine itself, rather than from added chemicals. This provides a significant advantage, as the stabilizing properties remain consistent over time, eliminating the need for complex or costly maintenance [7].

The frictional behavior of an SAE 52100 bearing steel disc sliding against an SAE 52100 bearing steel ball in the presence of Cesium formate CsFo, Potassium formate KFo, and their blends is shown in Figure 3 (a) and Figure 3 (b). The average CoF obtained with pure and mixed brines are illustrated in Table 3. It is notable that the lowest average CoF was reached with testing Potassium formate brine, where the average CoF was reduced to 0.095 and 0.093 in tests 1 and 2, respectively. Furthermore, a rise in the quantity of potassium in the mixture is accompanied by a reduction in the density and friction.

Table 3—Average CoF of CsFo, KFo and their associated blends

Samples	Density, SG	Test 1	Test 2	Average
CsFo	2.20	0.158	0.169	0.164
CsFo:KFo 75/25	2.05	0.138	0.128	0.133
CsFo:KFo 50/50	1.88	0.109	0.113	0.111
CsFo:KFo 25/75	1.70	0.092	0.098	0.093
KFo	1.57	0.095	0.093	0.094

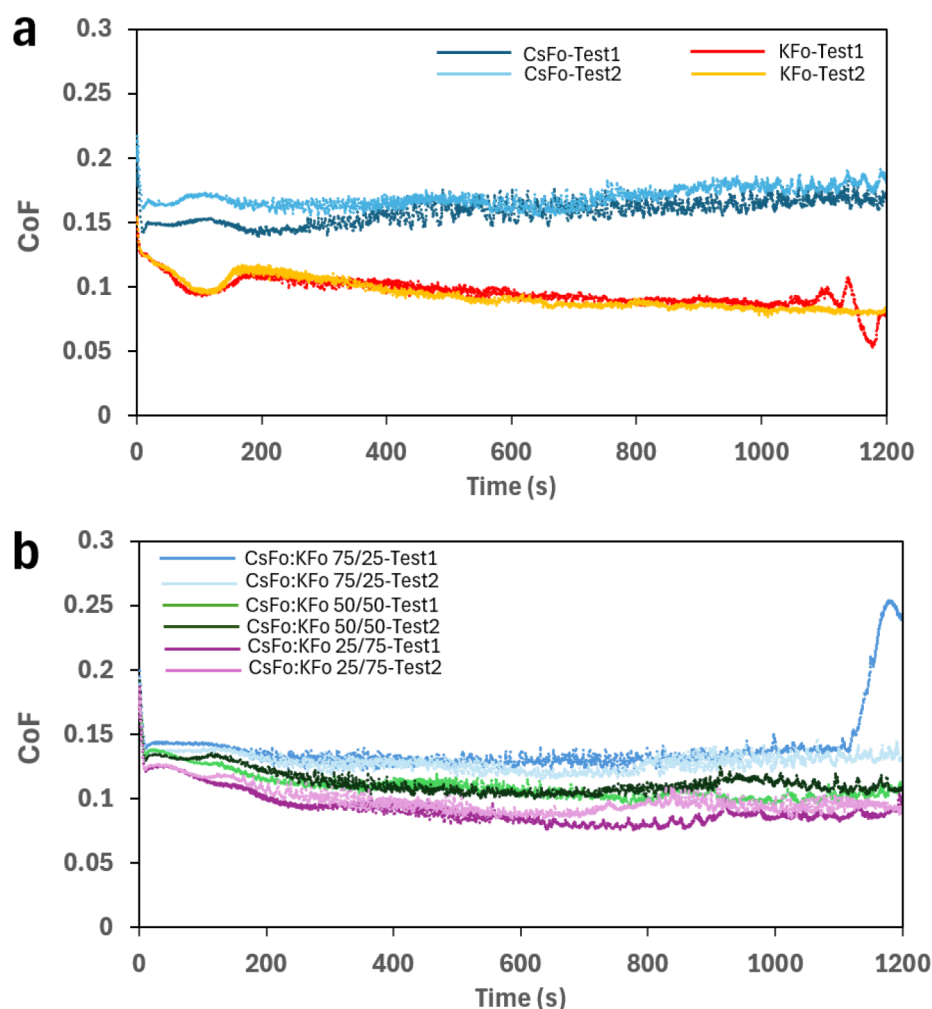


Figure 3—Variation of coefficient of friction with time for a) CsFo and KFo and b) CsFo/KFo blends.

The results of this experiment indicate that the positively charged ions of Potassium have been gradually adsorbed into the surface of the steel, and the interfacial tension (steel/brine) has been lowered as the Potassium ions rise, as indicated by the lower friction [8]. On the other hand, the observed rise in friction reported for Cesium underscores the importance of the type of salt, as Cesium exhibited poor adsorption onto the steel surface [9]. In addition, the carboxylate ion (RCOO^-) tends to aggregate at the interface and function as a hydrotropic molecule. The beneficial effect of this phenomenon may be stronger in KCOOH when compared to CsCOOH because KCOOH has a higher ionic strength than CsCOOH [10].

Potassium formate and Mn_3O_4 mixtures

As discussed in the previous section, potassium formate exhibits a low coefficient of friction (CoF) in steel-steel contact, along with a lower density. The next step was to investigate the effect of Mn_3O_4 , as a solid weighting material for potassium formate brine, on the coefficient of friction. Different mixtures of Potassium formate brines (KFo brine) with Mn_3O_4 and other solids like CaCO_3 and ilmenite were formulated and tested for friction as shown Table 4. No other additives like dispersant or viscosifier were added.

Table 4—Average coefficient of friction of KFo brine and KFo/Mn₃O₄ blends

Samples	KFo brine in gm	Mn ₃ O ₄ in gm	Density, SG	Test1	Test2	Average
KFo brine			1.57	0.095	0.093	0.094
KFo brine/ Mn ₃ O ₄ : 100/13	100	13	1.70	0.188	0.179	0.183
KFo brine/ Mn ₃ O ₄ : 100/29	100	29	1.85	0.184	0.180	0.182
KFo brine/ Mn ₃ O ₄ : 100/47	100	47	2.00	0.177	0.184	0.180
KFo brine/ Mn ₃ O ₄ : 100/67	100	67	2.15	0.190	0.188	0.189
KFo brine/ CaCO ₃ : 100/13	100	13	1.70	0.144	0.141	0.143
KFo brine/ ilmenite: 100/13	100	13	1.68	0.174	0.173	0.173

Table 4 shows that increasing the amount of Mn₃O₄ leads to a rise in friction. Unlike CsFo/KFo brines, which exhibited an almost linear correlation between the coefficient of friction (CoF) and brine density, no such correlation was observed for Mn₃O₄/KFo in the density range of 1.70–2.20 SG. Regardless of the amount of for Mn₃O₄ added, the CoF remained approximately 0.180.

Additional tests using other solids, such as ilmenite and CaCO₃ at 100/13 wt/wt%, yielded comparable friction values. This suggests that the increase in friction is not directly related to the chemical composition of the weighting material, but rather to the overall increase in solid content.

Interestingly, increasing the density of KFo brine—whether by adding CsFo brine or solid Mn₃O₄—resulted in a rise in CoF. Increasing the concentration of salts or solids in KFo brine raises its density by displacing water content. This reduction in water increases the fluid's internal resistance (viscosity), thereby elevating frictional forces and resulting in a higher coefficient of friction (CoF). It is important to note that these simple mixtures, even with dispersants, do not reflect the performance of fully formulated drilling fluids, which typically exhibit lower CoF values.

Effect of different lubricant additives on friction

Figure 4 illustrates the evolution of friction during two repetitions of steel-on-steel configuration tests conducted in pure KFo solution and in formulated Fluid 1. As shown, the tests demonstrated good repeatability. Consistent with the findings in Table 4, the addition of Mn₃O₄ to Fluid 1 resulted in an increase in friction.

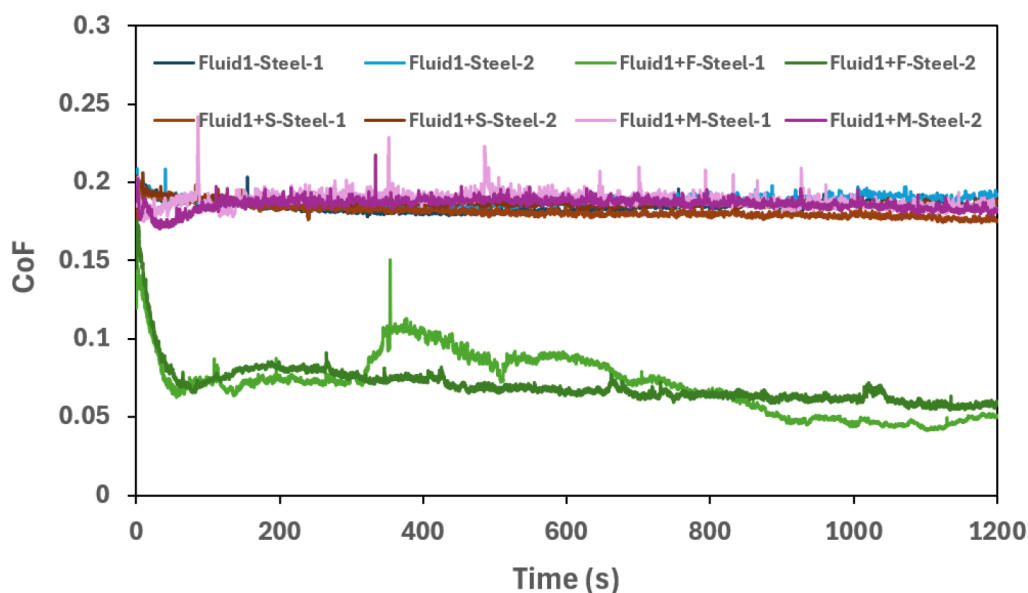


Figure 4—Friction evolution of the drilling fluid 1 with and without lubricating additives.

To reduce the coefficient of friction, three additives—F, M, and S—were introduced into Fluid 1 at a concentration of 3 vol% (as detailed in Table 2). The CoF results for both the base brine and the additive-enhanced fluids are presented in Figure 4. The data indicate that additives M and S had minimal impact on friction. However, the inclusion of 3 vol% of additive F led to a significant reduction in CoF, achieving values below 0.1.

Figure 5 shows the 3D images of the steel balls and the 2D images of the discs after performing tests in fluid 1 with and without additives.

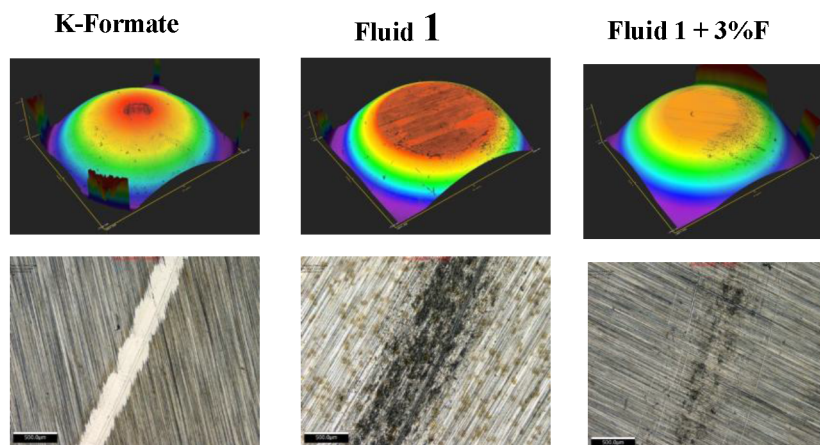


Figure 5—3D images of the steel balls (up) and 2D images of the steel discs (bottom) after sliding test.

As Figures 4 and 5 reveal, additive F functions simultaneously as both a friction modifier and an anti-wear additive, whereas adding 3 vol% of additives M and S did not alter either the frictional or wear behavior of drilling fluid 1.

Previous research on commercial anionic surfactants, such as Alpha Olefin Sulfonate (AOS), has demonstrated that the presence of potassium formate brine can effectively reduce interfacial tension [11]. This phenomenon is attributed to the presence of electrolytes, which enhance the electrostatic interactions between similarly charged ionic head groups, leading to a more compact arrangement of surfactant molecules at the interface. Consequently, this tighter packing results in a decrease in interfacial tension [8].

Furthermore, a study investigating the adsorption behavior of sodium dodecyl sulfate (SDS) on a stainless-steel surface in an aqueous solution revealed a similar correlation. The coefficient of friction decreased proportionally with interfacial tension until the SDS concentration reached the critical micelle concentration [12]. This behavior appears to be analogous to that of lubricant F, which also functions as an anionic substance in aqueous drilling fluid.

The critical micelle concentrations (CMCs) of non-ionic surfactants are generally lower than those of ionic surfactants in the presence of salts [13]. Unlike ionic surfactants, non-ionic surfactants do not exhibit electrostatic interactions that contribute to interfacial tension reduction. Previous studies have shown that non-ionic surfactants, such as Coco Glucoside, can lead to a slight increase in interfacial tension in salt-containing solutions, particularly at higher concentrations. This suggests that the presence of salt may negatively affect the surfactant's efficiency, leading to a salting-out effect [8]. A similar behavior has been observed for lubricant S, which also functions as a non-ionic surfactant in brine systems.

Lubricant M is derived from Carboxylic Acid (C12-lauric acid) which is used as a friction modifier in conventional lubricants [14]. In a separate study, the influence of polar head group type was analyzed—specifically the distinction between additive M and lauric acid—on the surface adsorption and frictional behavior of Organic Friction Modifiers (OFMs) [15]. This investigation aimed to elucidate how variations in molecular structure affect interfacial interactions and frictional and wear performance. The results indicated that although lubricant M exhibited a higher adsorption mass compared to lauric acid, the viscoelastic properties of the layer formed by additive M were inferior to those of the layer formed by lauric acid. This suggests that while additive M has a greater surface affinity, the structural integrity and mechanical resilience of the adsorbed film are more favorable in the case of lauric acid, potentially influencing its overall frictional and wear performance [15]

Sliding tests in steel-stone configuration. Figure 6 illustrates the variation in the (CoF) over time for tests conducted using a steel/stone contact configuration, in comparison to those performed with a steel/steel configuration. These experiments were carried out in fluid 1, both with and without the addition of additive F.

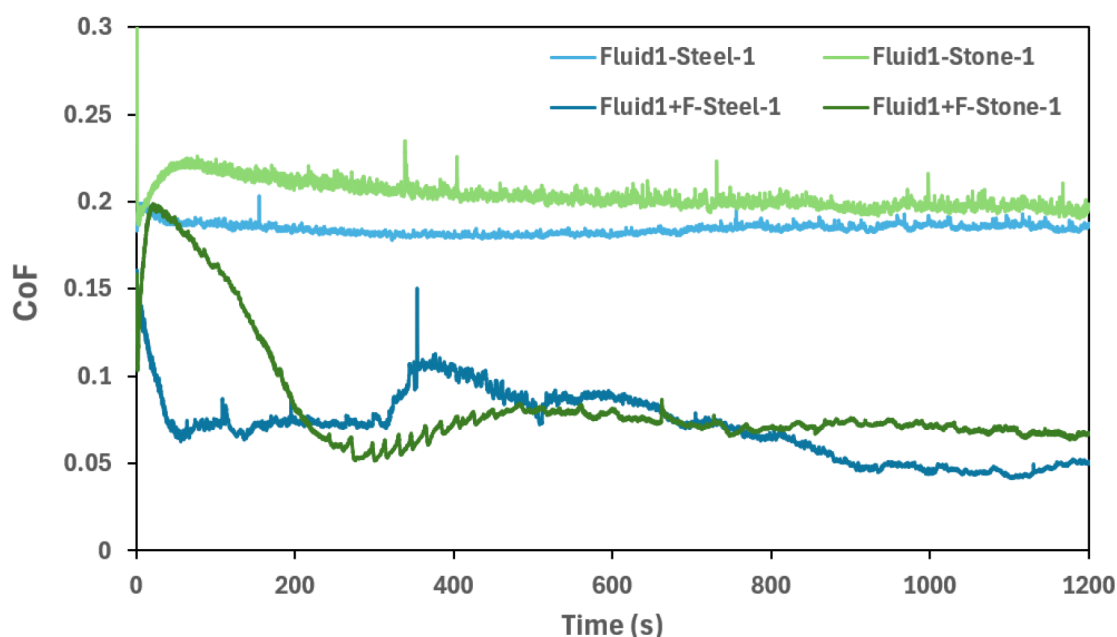


Figure 6—CoF evolution in tests performed in steel-steel (blue) and steel-stone (green) configurations with and without additive F.

The highest friction was found for the tests conducted in fluid 1 without any lubricant additive. In the steel-stone configuration, the (CoF) exhibits higher initial values at the beginning of the test. However, as the test progresses, the CoF gradually decreases and converges to the values observed in the steel-steel configuration towards the end of the test.

Friction is, as expected, lower for the tests conducted in fluid 1 with lubricant additive. In the case of the steel-stone configuration exhibited a prolonged running-in period characterized by high friction values compared to the steel-steel configuration during the first 200 seconds of testing, both configurations reached a stabilized (CoF) below 0.1 until the conclusion of the tests.

Conducting tribological tests under a more representative testing configuration using the steel-stone contact pair demonstrated that, despite the extended running-in period observed in the steel-stone tests, additive F remained effective in interaction with the shale stone surface. This effectiveness led to a reduced CoF value after the running-in period.

Figure 7 shows the (CoF) evolution with time for the steel-stone configuration tested with drilling fluid 2, both in the presence and absence of additive F. The formulation containing the additive includes 3 vol% of additive F, consistent with the concentrations used in the preceding sections.

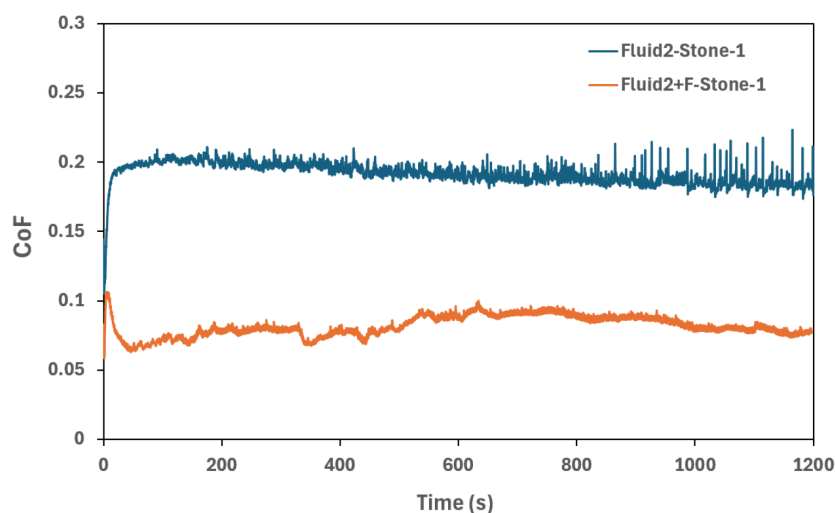


Figure 7—CoF evolution for steel-stone configuration in drilling fluid 2 with and without additive F.

Figure 7 demonstrates that the incorporation of additive F in drilling fluid 2 yields a comparable reduction in CoF as observed in fluid 1, decreasing to values below 0.1. This indicates that, despite drilling fluid 2 having a higher concentration of Mn_3O_4 particles and a greater density compared to fluid 1, additive F is still able to adsorb to the surface, thereby successfully functioning as a friction modifier.

Figure 8 presents the evolution of the (CoF) with time for tests conducted using stone/steel configuration in fluids 1, 2 and 3, each containing 3 vol.% of additive F. As detailed in Table 2, fluid 1 represents the least formulated composition, exhibiting the lowest density, while drilling fluids 2 and 3 are more advanced formulations more closely resembling the target application. Figure 8 reveals distinct running-in behaviors in the first 10 minutes of testing: the shortest running-in period is observed for drilling fluid 2 containing additive F, whereas the longest running-in period corresponds to drilling fluid 3 with the same additive. Following the running-in period, the CoF values for all tests stabilize at or below approximately 0.1 and remain relatively constant throughout the test duration. The average CoF values recorded over the final 200 seconds of testing were 0.07 for fluid 1, 0.08 for fluid 2, and 0.09 for drilling fluid 3. These values are comparable to, if not lower than, the coefficient of friction typically observed in most lubricious non-aqueous drilling fluids.

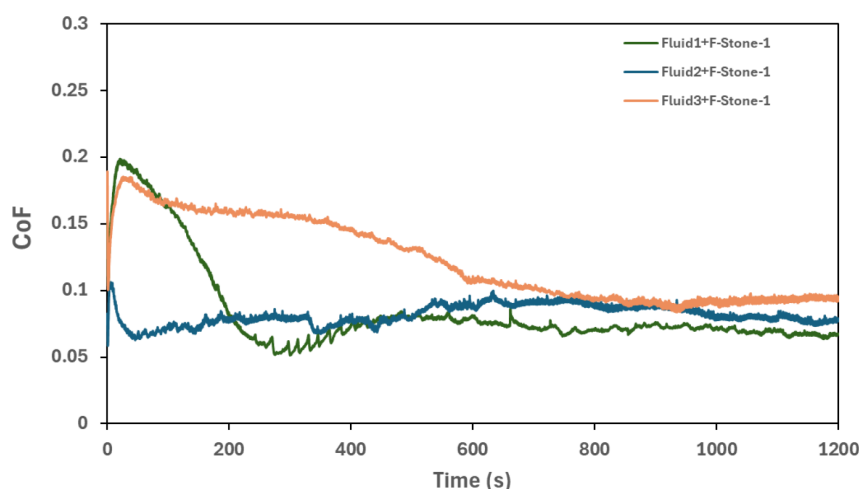


Figure 8—Friction evolution with time in steel-stone configurations in drilling fluid 1, 2, and 3 with additive F.

The observed variation in running-in behavior suggests different surface interaction dynamics during the initial stages of sliding. Specifically, drilling fluid 2, which exhibited the shortest running-in period, may possess an optimized balance of base formulation and additive F concentration that facilitates rapid surface adsorption and film formation. In contrast, the prolonged running-in period observed with drilling fluid 3 may be attributed to its higher viscosity or increased concentration of solid components (e.g., Mn_3O_4), which could delay the establishment of a stable adsorbed film. Despite the differences during the running-in phase, all three drilling fluids demonstrated stable friction values below or around 0.1 during the steady-state phase, indicating effective lubrication across the board. Notably, the average CoF values during the final 200 seconds of testing revealed a clear trend, i.e. increasing friction with increasing formulation complexity / density (0.07 for drilling fluid 1, 0.08 for drilling fluid 2, and 0.09 for drilling fluid 3).

The compositional differences among the three drilling fluids provide critical insights into the observed frictional performance. Fluid 1, the simplest formulation, primarily comprises potassium formate brine, Mn_3O_4 , and minor quantities of water, formic acid, and defoamer. Its minimalistic composition and relatively low solid content (e.g., 154.2 g of Mn_3O_4) are consistent with its low average coefficient of friction (0.07) and rapid stabilization during testing. The absence of viscosifier and fluid loss additives in fluid 1 likely results in reduced film thickness and interfacial resistance, promoting a lower CoF under boundary or mixed lubrication regimes.

In contrast, drilling fluid 2 includes a broader range of additives like xanthan gum, starch, dispersant, and a higher concentration of Mn_3O_4 (226.7 g). These components enhance rheological properties and particle suspension, mimicking operational drilling fluids more closely. The more complex matrix may lead to thicker adsorbed or delayed surface conformity during the running-in phase, although it still achieves a moderate friction (0.08), indicating a balance between enhanced adsorbed film formation and increased shear resistance.

Drilling fluid 3 represents an HPHT formulation, incorporating multiple polymeric additives (synthetic and cellulose-based), calcium carbonate, and the highest Mn_3O_4 content (385.6 g). While these components are essential for enhancing fluid performance, they also increase viscosity and the concentration of load-bearing solids at the drilling interface—the zone where the drill bit, drilling fluid, and formation interact. This is consistent with the longest running-in time and highest steady-state friction (0.09) observed. The presence of multiple polymers could also introduce complex molecular interactions at the contact interface, further contributing to frictional resistance.

Optimizing the additive content using adsorption studies

Quartz Crystal Microbalance with Dissipation (QCM-D) is a highly sensitive, real-time analytical technique used to monitor mass changes and viscoelastic properties at surfaces. Based on the piezoelectric effect, QCM-D measures shifts in the resonance frequency (Δf) and energy dissipation (ΔD) of a quartz crystal sensor as material adsorbs to or desorbs from its surface. The frequency change correlates with mass deposition, while dissipation reflects the rigidity or softness of the adsorbed layer. This dual-parameter capability makes QCM-D particularly valuable for studying thin films, biomolecular interactions, and adsorption kinetics. In the context of additive optimization, QCM-D enables precise tracking of how varying concentrations influence adsorption behavior, layer structure, and film stability at the molecular level.

To model the surfaces of the steel and stone contacting pairs examined in the preceding chapters, coated quartz sensors were employed. Specifically, 5 MHz AT-cut quartz crystals coated with iron (Fe) were used to simulate the steel surface. In the case of the stone counterpart, a detailed X-ray diffraction (XRD) analysis of the shale stone utilized in the sliding tests identified kaolinite as the dominant mineral phase. Accordingly, kaolinite-coated quartz sensors were prepared to represent the stone surface in the QCM-D experiments.

For the water-based solutions, pure distilled water was selected as the base fluid for dispersing additive F. This choice was made due to the high sensitivity of the QCM-D technique to the presence of rigid particles and elevated viscosities, which can complicate data interpretation. To maintain a simplified and controlled experimental setup, aqueous solutions containing 0.2, 0.5, 1.5, and 3 vol.% of additive F were prepared. QCM-D measurements were performed sequentially in the following order to ensure consistent rinsing and to monitor adsorption/desorption behavior: distilled water $\rightarrow$ 0.2 vol.% F $\rightarrow$ distilled water $\rightarrow$ 0.5 vol.% F $\rightarrow$ distilled water $\rightarrow$ 1.5 vol.% F $\rightarrow$ distilled water $\rightarrow$ 3 vol.% F $\rightarrow$ final rinse with distilled water.

Figure 9 illustrates the evolution of the resonance frequency (Δf) during the QCM measurements on the surface of Fe-coated and Kaolinite-coated quartz crystals. Both frequency shift (Δf) profiles reveal clear, concentration-dependent adsorption behavior of additive F on the modeled contacting surfaces. In each case, a stepwise increase in mass adsorption is indicated by progressively larger negative frequency shifts upon exposure to higher concentrations of the additive. The reversibility of adsorption is assessed by the return to baseline or partial recovery during rinsing steps with distilled water. Additive F exhibits strong and stable interaction with iron, particularly at concentrations ≥ 0.5 vol.%, indicating effective surface coverage and potential for friction and wear improvement. The kaolinite-coated sensor shows significantly larger frequency shifts, which can be due to the higher real surface area, porosity and possibly hydration reaction of Kaolinite with water.

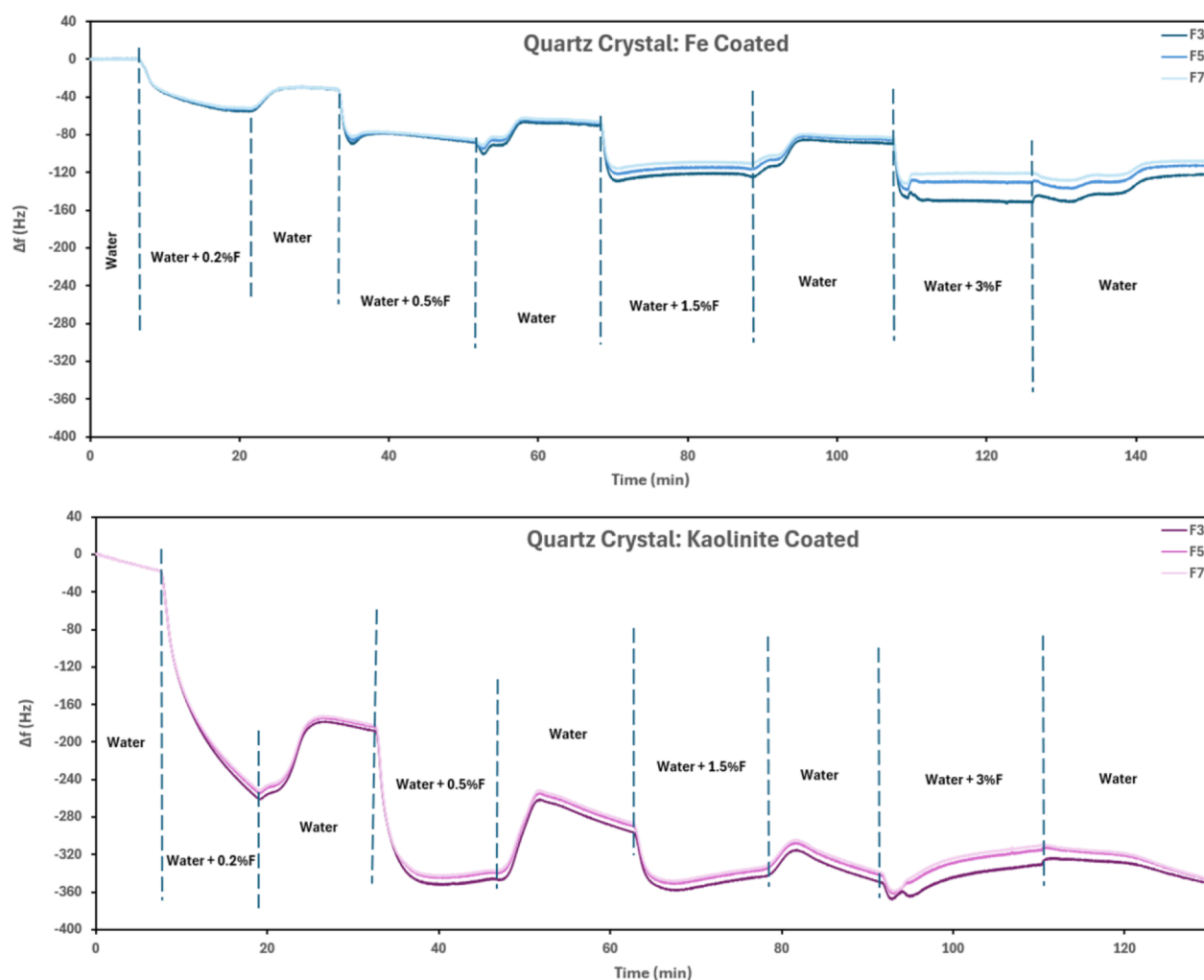


Figure 9—The resonance frequency evolution on Quartz sensors coated with iron and Kaolinite.

As shown in both graphs of Figure 10, the concentration of 3 vol.% F appears to correspond to a saturation point, beyond which no significant additional adsorption is observed. Furthermore, subsequent rinsing with distilled water does not result in noticeable desorption, indicating strong and stable binding of the additive to both surfaces at this concentration. To investigate the effect of additive F concentration on the coefficient of friction (CoF), sliding tests were conducted using a steel-steel contact configuration. Fluid 1, containing 0.5, 1.5, and 3 vol.% of additive F, was tested and the resulting CoF values are presented in Figure 10.

As shown in Figure 10, increasing the concentration of additive F from 0.5 to 1.5 and 3 vol.% results in a progressive reduction in both the running-in duration and the stabilized coefficient of friction (CoF). In particular, the drilling fluid containing 3 vol.% F exhibits the shortest running-in period—approximately 50 seconds—and maintains a stable CoF around 0.06 throughout the test. This observation aligns with the QCM-D adsorption results, which indicated surface saturation at 3 vol.% F, suggesting that optimal surface coverage contributes to enhanced friction-reducing performance.

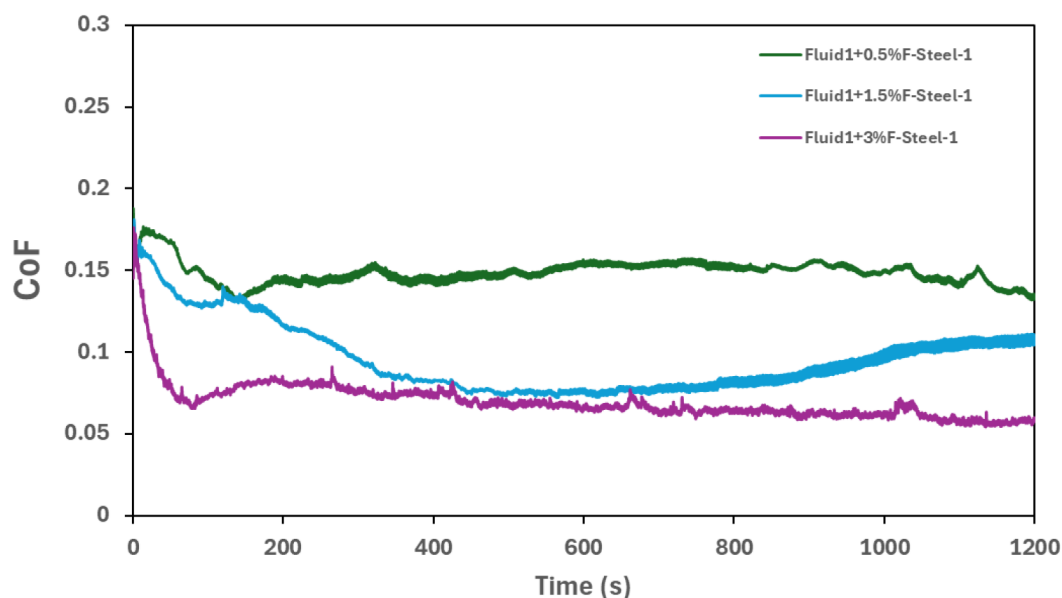


Figure 10—CoF evolution in steel/steel configurations in fluid containing different concentrations of lubricant F.

Comparing friction measurements by Tribometer and oilfield lubricity tester

We conducted a comparative study between a tribometer and a commonly used oilfield lubricity tester, such as the OFITE device. In the standard lubricity coefficient test, 150 in-lb of force—equivalent to approximately 5,000 to 10,000 PSI of pressure on the intermediate fluid—is applied between two hardened steel surfaces: a stationary block and a rotating ring at 60 RPM.

Under comparable test conditions (load, speed, and temperature), the results obtained were quite similar, as shown in Table 5. The OFITE lubricity tester consistently produced slightly higher CoF values. It is important to note, however, that the OFITE tester provides only a single-point measurement, and significant variability between readings can occur.

Table 5—Coefficient of friction of KFo/Mn₃O₄ fluids measured by tribometer and lubricity tester

Fluid	Tribometer	EP lubricity tester
Fluid 1 with 3vol.% additive F	0.06	0.094
Fluid 2 with 3vol.% additive F	0.07	0.09
Fluid 3 with 3vol.% additive F	0.10	0.13

Impact of lubricant on fluid performance

While the primary function of the lubricant additive is to enhance the lubricity of the drilling fluid, it is essential that it does not negatively impact other critical fluid properties such as viscosity, filtration control, or sag stability. As demonstrated in Table 6, the 1.90 SG Mn₃O₄/KFo drilling fluid containing the F-type lubricant maintained its properties after hot rolling at 135 °C for 16 hours. This indicates that the lubricant is thermally stable and does not adversely interact with other additives in the formulation. It is also noteworthy that the lubricant was introduced at the final stage of mixing, at a low shear rate, after all other additives had been incorporated.

Table 6—Properties of 1.90 SG KFo/Mn₃O₄ fluid with and without lubricant

Drilling Fluid	Ref fluid		Fluid with 1vol % lubricant F	
	gm	ml	gm	ml
Potassium Formate brine, 1.55 SG	472.1	304.6	472.1	304.6
HPHT fluid loss additive	3.6	2.4	3.6	2.4
PAC LV	2.0	1.3	2.0	1.3
Xathan gum	0.6	0.5	0.6	0.5
Sodium bicarbonate	0.75	0.3	0.75	0.3
Soda ash (sodium carbonate)	0.08	0.0	0.08	0.0
Dispersant	1.57	1.3	1.57	1.3
CaCO ₃ , 20μ	9.9	3.7	9.9	3.7
Mn ₃ O ₄	174.0	35.9	174.0	35.9
Lubricant F	0.0	0.0	3.6	3.5
Total	664.1	350.0	664.1	353.5
Static heat aging at 135°C for 16h	BSHA	ASHA	BSHA	ASHA
600 rpm	126	67	84	83
300 rpm	74	39	49	51
200 rpm	55	29	37	39
100 rpm	33	18	23	25
6 rpm	5	3	4	5
3 rpm	4	3	3	4
PV @ 120°F, cP (50 °C)	52	28	35	32
YP, lb/100ft ²	22	11	14	19
10 sec Gel, lb/100ft ²	4	3	5	5
10 min Gel, lb/100ft ²	23	3	19	4
Fluid loss at 135°C	15	11	15	12
Filter cake (mm)	1,5	2	1,5	2
Density g/ml before lubricity test	1,88	1,89	1,87	1,88
Coefficient of Friction (μ)	0,134	0,154	0,101	0,110
Supernatant liquid (ml)		ca 1		ca 3
Density Top		1,76		1,84
Density Bottom		1,89		1,91
Sag Factor		0,518		0,509

Conclusions

After testing different types of water-based drilling fluids and lubricant additives with Mn₃O₄/KFo, and based on the results obtained in this work, the following conclusions can be drawn:

- Potassium formate brine (KFo) provides the lowest (CoF) among the tested brines, achieving values as low as 0.093, indicating very good lubricity and surface interaction.

- Increasing the density of KFo brine—either by adding CsFo brine or incorporating solids such as Mn_3O_4 —results in a comparable increase in friction.
- The increase in friction from weighting materials like Mn_3O_4 , CaCO_3 , and ilmenite is attributed to solid content, rather than to specific chemistry.
- Among the lubricant additives tested, additive F significantly reduces both friction and wear, achieving coefficients of friction below 0.10.
- Additive F remains effective in both steel-steel and steel-stone configurations, even under higher-density formulations, maintaining friction below 0.10 after the running-in period.
- Increasing the formulation complexity and density of drilling fluids, increases the running-in period and the steady-state friction (from 0.07 to 0.09), but additive F maintains the lubrication efficiency.
- QCM-D analysis confirms strong, concentration-dependent adsorption of additive F on both iron and kaolinite surfaces, with the saturation point observed at 3 vol. %.
- Additive F shows irreversible or strongly bound adsorption, especially on kaolinite, supporting its effectiveness in shale-contact drilling scenarios.
- High-density Mn_3O_4 /KFo-based fluids, formulated with an appropriate lubricant, exhibit a low friction factor comparable to that of highly lubricious non-aqueous drilling fluids. This makes such fluid well suited for drilling long ERD wells.

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References

1. Zheng, Y., A. Amiri, and A.A. Polycarpou, Enhancements in the tribological performance of environmentally friendly water-based drilling fluids using additives. *Applied Surface Science*, 2020. **527**: p. 146822.
2. Caenn, R. and G.V. Chillingar, Drilling fluids: State of the art. *Journal of petroleum science and engineering*, 1996. **14**(3-4): p. 221–230.
3. Akpan, E.U., G.C. Enyi, and G.G. Nasr, Enhancing the performance of xanthan gum in water-based mud systems using an environmentally friendly biopolymer. *Journal of Petroleum Exploration and Production Technology*, 2020. **10**: p. 1933–1948.
4. Samavati, R., et al, The prospect of utilizing a cassava derivative (fufu) as a fluid loss agent in water based drilling muds. *International Journal of Chemical Engineering and Applications*, 2014. **5**(2): p. 161–169.
5. Khanmohammadi, H., W. Wijanarko, and N. Espallargas, Ionic liquids as additives in water-based lubricants: from surface adsorption to tribofilm formation. *Tribology Letters*, 2020. **68**(4): p. 130.
6. Van Oort, E., On the physical and chemical stability of shales. *Journal of Petroleum Science and Engineering*, 2003. **38**(3-4): p. 213–235.
7. Formate Technical Manual. 2019, SINOMINE Specialty Fluids.
8. Lezama, L.T., R. Rafati, and A.S. Haddad, The effect of formate brines on the emulsification of ionic, non-ionic, and enzymatic surfactants. *Chemical Engineering Science*, 2022. **263**: p. 118016.
9. Weissenborn, P.K. and R.J. Pugh, Surface tension of aqueous solutions of electrolytes: relationship with ion hydration, oxygen solubility, and bubble coalescence. *Journal of colloid and interface science*, 1996. **184**(2): p. 550–563.

10. Minofar, B., et al, Propensity of formate, acetate, benzoate, and phenolate for the aqueous solution/vapor interface: Surface tension measurements and molecular dynamics simulations. *The Journal of Physical Chemistry C*, 2007. **111**(23): p. 8242–8247.
11. Para, G., et al, Effect of electrolytes on surface tension of ionic surfactant solutions. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 2003. **222**(1-3): p. 213–222.
12. Zhang, J. and Y. Meng, Stick-slip friction of stainless steel in sodium dodecyl sulfate aqueous solution in the boundary lubrication regime. *Tribology Letters*, 2014. **56**: p. 543–552.
13. Miyagishi, S., K. Okada, and T. Asakawa, Salt effect on critical micelle concentrations of nonionic surfactants, N-acyl-N-methylglucamides (MEGA-n). *Journal of Colloid and Interface Science*, 2001. **238**(1): p. 91–95.
14. Bernat, S., S. Armada, and N. Espallargas, Friction mechanisms by carboxylic acids in aqueous lubricants. *Tribology Letters*, 2018. **66**: p. 1–15.
15. Marmorat, T., et al, Effect of the Polar Head Type on the Surface Adsorption and Tribofilm Formation of Organic Friction Modifiers in Water-Based Lubricants. *Langmuir*, 2024. **40**(15): p. 7920–7932.